



ASKING

smith_2025.pdf

Smith et al. (2025) · 31 pages · only this source is used

Change scope ▾

How did the authors measure cell temperature?

Adam · just now · 3 pages used

ANSWER

Cell temperature was measured with a **type-K thermocouple** bonded to the rear of each cell **p.8** and cross-checked against the plate temperature, with a stated difference below 1 K **p.9**.

Where it says so

Page 8 · 2.3 Instrumentation

"Each cell carries a type-K thermocouple bonded to its rear contact with thermally conductive epoxy."

Open page

Page 9 · 2.4 Procedure

"Plate and cell temperatures agreed to within 1 K at every step before a measurement was taken."

Open page

Not found in this source: the thermocouple's accuracy class. You could ask the workspace, or check datasheet_cell.pdf. [Ask all sources instead](#)

Scope

This source

Selected

All 12

Switching scope re-runs the question. The scope always stays visible above the answer.

SELECTED SOURCES · example

☒

instructions.pdf

☒

lecture_05.pdf

☒

measurements.xlsx

☐

notes.docx

☐

sample_data.csv — processing

3 of 12 selected

Select all ready · Clear

Sources still **processing** can't be selected yet — sample_data.csv is at 64%.

[illegible]

Compare sources

Comparing

 smith_2025.pdf ×

 lee_2024.pdf ×

 lecture_05.pdf ×

+ Add source

SOURCE	 smith_2025.pdf Smith et al. (2025) · 31 pp. Open	 lee_2024.pdf Lee et al. (2024) · 18 pp. Open	 lecture_05.pdf Lecture 5 · 24 slides Open
Research question	How does cell temperature change the efficiency of monocrystalline silicon cells under controlled illumination? p.2	Does the efficiency–temperature relation hold for heterojunction modules at high irradiance? p.3	Why does the open–circuit voltage of a diode–based cell decrease with temperature? sl.4
Method	Thermostatic plate, 25–75 °C, solar simulator at 1000 W/m². p.8	Outdoor field measurements, 15–70 °C, two module types, three months. p.5	Derivation from the ideal diode equation. No measurements. sl.6
Dataset	5 cells · 12 temperature steps · 3 repeats each. p.9	Hourly data · 2 module types · about 2,100 hours. p.6	None — theory only.
Main finding	Efficiency falls $\approx 0.41\%$ /K; the Voc drop dominates the loss. p.14	Heterojunction modules lose $\approx 0.27\%$ /K — less than the PERC reference. p.9	$dVoc/dT \approx -2.2\text{ mV/K}$ for silicon at 300 K. p.22
Limitations	Single cell architecture; indoor simulator only. p.17	Irradiance not controlled; sensor drift not quantified. p.12	Idealised single–diode model; ignores series resistance. sl.18
Relevant evidence	“...conversion efficiency fell by 0.41 % per kelvin...” — cited in Laboratory Report [1].	“...a coefficient of -0.27% /K...” — saved, not cited. p.9	“... $\approx -2.2\text{ mV/K}$...” — cited in Laboratory Report [2].
Contradictions	None with lecture_05.pdf. Narrower temperature range than lee_2024.pdf.	Coefficient is lower than in smith_2025.pdf (0.27 vs 0.41 %/K). Different architecture and method — see contradiction view.	None.

🌟 SUGGESTED SUMMARY · draft for your report

Laboratory and simulator data agree that silicon cell efficiency falls by about 0.4 % per kelvin [1](#) [3](#). Field data for heterojunction modules suggest a smaller coefficient of about 0.27 % per kelvin [2](#), so the coefficient depends on cell architecture.

⚠️ 1 difference worth a look

The temperature coefficient differs between smith_2025.pdf and lee_2024.pdf.

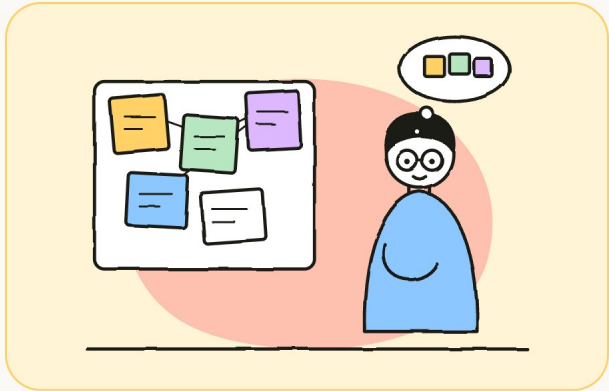
[Open contradiction view](#)

CLAIM · from 2. Theoretical Background

Partly supported · needs qualifying

“Increased temperature always reduces conversion efficiency.”

Two sources support the general trend. One source reports a narrower case where the trend does not hold. That's useful research information — not an error.



✓ Supporting · 2 sources

1 smith_2025.pdf · page 14

“...conversion efficiency fell by 0.41 % per kelvin over the 25–75 °C range...”

Indoor simulator · 1000 W/m² · crystalline Si

2 lecture_05.pdf · page 22

“ $dV_{oc}/dT \approx -2.2 \text{ mV/K}$ — efficiency decreases with temperature.”

Derivation · ideal single-diode model

⚠ Qualifying · 1 source

3 lee_2024.pdf · page 9

“At irradiance below 200 W/m², efficiency rose slightly between 15 and 30 °C before falling again.”

Heterojunction modules · outdoor measurement

Handle with care. Single source, field conditions, sensor drift not quantified.

Why the sources differ

⚡ Different temperature range

Supporting: 25–75 °C

Qualifying: 15–30 °C (the rise)

📦 Different cell architecture

Supporting: Monocrystalline Si

Qualifying: Heterojunction

🧪 Different measurement method

Supporting: Controlled simulator

Qualifying: Outdoor, uncontrolled

★ SUGGESTED WORDING · replaces the claim in your report

Efficiency generally decreases as temperature rises, by about 0.4 % per kelvin for monocrystalline silicon between 25 and 75 °C **1** **2**. The effect can be weaker, or briefly reversed at low irradiance, for other cell architectures **3**.

📄 Insert into document

Keep as open question